Literature

Wiki *Enhanced dissipation for passive scalars*

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1 Reviews

Reviews (3).

- A. L. Mazzucato, Y. Feng and C. Nobili, *Enhanced dissipation by advection and applications to PDEs*, Applied Mathematical Modelling **150** (2026), Part A, 116310, arXiv:2501.17695.
- M. Coti Zelati, G. Crippa, G. Iyer and A. L. Mazzucato, *Mixing in incompressible flows: transport, dissipation, and their interplay*, Notices of the American Mathematical Society **71** (2024), 593–604, arXiv:2308.00358.
- T. Gallay, *Estimations pseudo-spectrales et stabilité des tourbillons plans (d'après Te Li, Dongyi Wei et Zhifei Zhang)*, Séminaire Bourbaki, exposé n°1167, Astérisque (2022), doi:10.24033/ast.1156.

2 Deterministic enhanced dissipation

Shear, radial and other symmetric flows (19).

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- M. Coti Zelati and T. Gallay, *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows*, Journal of the London Mathematical Society **108** (2023), 1358–1392, arXiv:2108.11192.
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- M. Coti Zelati and T. D. Drivas, *A stochastic approach to enhanced diffusion*, Annali della Scuola Normale Superiore di Pisa, Classe di Scienze **22** (2021), 811–834, arXiv:1911.09995.
- M. Coti Zelati, *Stable mixing estimates in the infinite Péclet number limit*, Journal of Functional Analysis **279** (2020), 108562, arXiv:1909.01310.
- M. Coti Zelati and M. Dolce, *Separation of time-scales in drift-diffusion equations on R^2*, Journal de Mathématiques Pures et Appliquées **142** (2020), 58–75, arXiv:1907.04012.
- M. Beck, O. Chaudhary and C. E. Wayne, *Rigorous justification of Taylor dispersion via center manifolds and hypocoercivity*, Archive for Rational Mechanics and Analysis **235** (2020), 1105–1149, arXiv:1804.06916.
- J. Bedrossian and M. Coti Zelati, *Enhanced dissipation, hypoellipticity, and anomalous small noise inviscid limits in shear flows*, Archive for Rational Mechanics and Analysis **224** (2017), 1161–1204, arXiv:1510.08098.
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- A. Kolmogorov, *Zufällige Bewegungen (Zur Theorie der Brownschen Bewegung)*, Annals of Mathematics (2) **35** (1934), 116–117, doi:10.2307/1968123.

- W. Thomson (Lord Kelvin), *Stability of fluid motion: rectilineal motion of viscous fluid between two parallel planes*, Philosophical Magazine (5) **24** (1887), 188–196, doi:10.1080/14786448708628078.

Other smooth flows (4).

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- E. Bruè, M. Coti Zelati and E. Marconi, *Enhanced dissipation for two-dimensional Hamiltonian flows*, Archive for Rational Mechanics and Analysis **248** (2024), Article 84, arXiv:2211.14057.
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- T. M. Elgindi and K. Liss, *Norm growth, non-uniqueness, and anomalous dissipation in passive scalars*, Archive for Rational Mechanics and Analysis **248** (2024), arXiv:2309.08576.
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3 Stochastic enhanced dissipation

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- J. Bedrossian, A. Blumenthal and S. Punshon-Smith, *Almost-sure enhanced dissipation and uniform-in-diffusivity exponential mixing for advection-diffusion by stochastic Navier-Stokes*, Probability Theory and Related Fields **179** (2021), 777–834, arXiv:1911.01561.

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- M. Coti Zelati, M. Hairer and D. Villringer, *A Stochastic RAGE Theorem and Enhanced Dissipation for Transport Noise*, preprint (2025), arXiv:2507.11422.

4 Deterministic mixing

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5 Stochastic mixing

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